

A factorial-scale barrier even for error-dependent relabeling averages

Mathematical proof reviewed independently; not formally verified

30 September 2026

Let $V = \text{Lie}_n$ be the real multilinear degree- n free Lie representation on n distinct generators. It has dimension $(n-1)!$. For $E \in V$ let $q(E)$ be the least support size of a probability measure μ on S_n such that $\sum_g \mu(g)gE = 0$. Such measures always exist, by uniform averaging. Unlike a universal annihilator, the measure here may depend on E .

Theorem 1. *For $n \geq 7$ put*

$$Q_n = \left\lfloor \sqrt{\frac{n!}{2n^5}} \right\rfloor.$$

If $Q_n \geq 1$, the set $\{E \in V : q(E) \leq Q_n\}$ is closed and has Lebesgue measure zero. Its complement contains an element with integral coordinates in the standard fixed-last-letter bracket basis.

In particular, for every sufficiently large n , there is a positive word w with equal letter counts, fair on every subset of at most $n-1$ letters, such that cancelling its top-degree defect by positive averaging of relabeled copies requires more than Q_n distinct relabelings. Thus the worst-case support requirement is

$$\exp((1/2 + o(1))n \log n).$$

Lemma 2 (Rank of a positive relabeling average). *For $n \geq 7$, if μ is supported on at most q permutations, then*

$$\text{rank}(T_\mu|_V) \geq \frac{n!/q - 2}{n^5}, \quad T_\mu = \sum_g \mu(g)g.$$

Proof. On the regular representation of S_n , every group element is orthogonal, so $\|T_\mu\|_{\text{op}} \leq 1$. Its squared Frobenius norm is $n! \sum_g \mu(g)^2 \geq n!/q$. Since each singular value is at most one, its regular-representation rank is at least $n!/q$.

For an irreducible complex representation S^λ , let d_λ be its dimension and m_λ its multiplicity in Lie_n . Klyachko's theorem gives $m_\lambda \geq 1$ for every shape except trivial and sign when $n \geq 7$. Moreover,

$$m_\lambda \geq d_\lambda/n^5 \quad \text{for every such shape.} \tag{1}$$

Indeed, this is immediate when $d_\lambda \leq n^5$; when $d_\lambda > n^5$, Swanson [1, Theorem 1.9], together with the Lie multiplicity interpretation, gives $m_\lambda/d_\lambda > 1/n - 1/n^2 \geq 1/n^5$.

If r_λ is the rank of $\sum_g \mu(g)\rho_\lambda(g)$, complexification and semisimplicity give

$$\text{rank}_{\text{reg}} T_\mu = \sum_\lambda d_\lambda r_\lambda, \quad \text{rank}_V T_\mu = \sum_\lambda m_\lambda r_\lambda.$$

The two omitted one-dimensional representations contribute at most two to the first sum. Inequality (1) proves the result. $\square$

Lemma 3 (A parameterized-kernel observation). *Let $A(t)$ be a linear operator on $\mathbb{R}^d$ whose entries depend linearly on t in a compact simplex of dimension at most $q - 1$. Suppose $\text{rank } A(t) \geq q$ throughout the simplex. Then*

$$\mathcal{B} = \{x : \text{some } t \text{ satisfies } A(t)x = 0\}$$

is closed and has Lebesgue measure zero.

Proof. Closedness follows by taking a convergent subsequence of parameters in the compact simplex whenever $x_j \rightarrow x$ and $A(t_j)x_j = 0$. For the measure statement, cover the parameter simplex by the finitely many sets where a specified $q \times q$ minor is nonzero. On each such set, q of the coordinates of a kernel vector are rational, hence smooth, functions of the parameters and the remaining $d - q$ coordinates. Ignoring any remaining equations only enlarges the resulting set. The total number of free real parameters is at most $(q - 1) + (d - q) = d - 1$. Restricting to countably many bounded pieces away from a zero denominator makes each parameterization Lipschitz, and a Lipschitz image of a subset of $\mathbb{R}^{d-1}$ has d -dimensional measure zero. The finite union of charts has the same property. $\square$

Proof of Theorem 1. Put $q = Q_n \geq 1$. Then $n! \geq 2n^5q^2 \geq n^5q^2 + 2q$, so Lemma 2 gives $\text{rank}(T_\mu|_V) \geq q$ for every measure supported on at most q permutations. For each fixed support of size q , its probability measures form a compact $(q - 1)$ -simplex. Lemma 3 shows that the errors it can cancel form a closed null set. There are finitely many supports; smaller supports are included by allowing zero weights. Their union is therefore closed and null. Its nonempty open complement contains a rational point in the rational bracket basis. Multiplication by a common denominator preserves membership in this complement, giving an integral error E with $q(E) > Q_n$.

We now realize this error by an actual positive word. In the reduced nilpotent group, every degree- n bracket is central and its group commutator is exactly its exponential. Thus an integral linear combination E of the fixed-last-letter brackets gives a signed word with signature $\exp(E) = 1 + E$. The positive fraction-clearing argument in `../positive_saturation/eventual_primorial_saturation.tex` then supplies an equal-count fair signature $T_L = \exp(LH)$, where $H = X_1 + \cdots + X_n$, such that $\exp(E)T_L$ is represented by a positive word w . Explicitly, write the signed element as uv^{-1} with positive u, v ; complete v to a fair positive word $vc = T_L$, and use uc . A further fair factor can make $L > 0$ if necessary. Centrality gives

$$\log S(w) = LH + E.$$

Hence all restrictions to fewer than n letters are fair and every letter occurs exactly L times.

For relabeled copies of w , the degree- n defects are gE . Their positive weighted average vanishes exactly when its normalized weights define a measure annihilating E , which needs more than Q_n relabelings. If integer copies and common integer blowups are used, a blowup by a multiplies the defect by $a^n > 0$; these still produce positive weights and cannot circumvent the support bound. $\square$

Scope. This is a worst-case obstruction for cancelling the final defect of an arbitrary almost-fair seed by relabeling. It does not show that every seed needs many relabelings, does not rule out exploiting particular seed structure, and gives no lower bound on the optimal size of a fair family constructed by other means. The representation estimates are classical; historical novelty of this application has not been audited.

References

- [1] Joshua P. Swanson, *On the existence of tableaux with given modular major index*, Algebraic Combinatorics 1 (2018), 3–21, doi:10.5802/alco.4.